ProveIt research contribution

Direct quadratic packing for a ten-integer consistency equation

A polynomial of exact degree 237 520 456 in 782 operations

AI-assisted research prepared for Vladimir Reshetnikov
23 October 2026

Abstract

We replace the source coding in the repository's ten-integer Diophantine representation of ZFC inconsistency. The replacement encodes the 22 quadratic parent equations directly, using 32 Sidon positions and 22 independently tested target digits. It never forms the quartic sum of squares before packing. The resulting completely fixed polynomial

$$F_{10}(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}[f, g, h, k, l, w, x, y, z, m]$$

has exact degree 237 520 456 and a division-free circuit with 782 binary arithmetic operations. On the inherited, unchanged K215/compiler coefficient tuple, ordinary mathematics formalizable in ZFC proves

$$\big(\nexists\, \mathbf{v} \in \mathbb{Z}^{10} : F_{10}(\mathbf{v}) = 0\big) \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

Compared with package 31 at the pinned repository revision, the number and domain of unknowns are unchanged, the degree falls by a factor exceeding $2.47 \cdot 10^{18}$, and the circuit loses 660 operations.

The new ingredient is a proved source reduction for quadratic systems. Sun's Lucas and binomial bridges, including his 2026 preprint, and the repository's modified three-square finalizer are retained with explicit attribution. The archive contains the polynomial circuit, all fixed coefficient recipes and payloads, the literal machine, exact algebraic audits, independent circuit checks, and this article. This is not an end-to-end proof-assistant certificate or a claim of global minimality.

Metric	Package 31	This construction
Unrestricted integer unknowns	10	10
Exact degree	$126496 \cdot 5^{31} + 27064$	237 520 456
Binary $+, -, \times$ operations	1,442	782
Native coefficient tuple	121×8801	unchanged

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1 The result and the precise comparison

The reference tree is Vladimir Reshetnikov's ProveIt repository at commit

`22e75dd6b87b1b199bb08c778d7cf075ae5aae08`.

The comparison is with the current collection and, specifically, its `31-sun10` package, rather than an earlier superseded wave [1, 2]. The copied collection description is `upstream/COMPARISON_AT_PIN.md`.

Fix the repository's effective proof calculus for ZFC. In this article

$$\mathrm{Con(ZFC)} \ :\Longleftrightarrow \ \neg\exists p \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)$$

means absence of a finite refutation in that calculus. This is the sentence calibrated by the endpoint `BProvZFCax_s[]Form.fBot` in `KEndpoint215.lean`. We use $\mathbb{N} = \{0, 1, 2, \ldots\}$; positive integers are denoted by $\mathbb{Z}_{>0}$.

Theorem 1.1 (Main construction)**.** *The definitions in Sections 2–5, with the fixed data supplied in the archive, determine one polynomial F_{10} over the integers in exactly the ten unknowns*

$$(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

Its exact total degree is 237 520 456*. The supplied circuit has 221 additions, 25 subtractions, and 536 multiplications. Composing the new reduction with the inherited K215/native-compiler equivalence and the cited Lucas theorems gives a proof formalizable in ZFC of*

$$\exists \mathbf{v} \in \mathbb{Z}^{10} \ F_{10}(\mathbf{v}) = 0 \quad \longleftrightarrow \quad \exists p \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot). \tag{1.1}$$

Consequently ZFC proves that absence of an integer zero is equivalent to Con(ZFC)*.*

All the auxiliary expressions below are polynomial abbreviations. They introduce no further existential unknowns. All exponents, finite sum limits, and coefficients are fixed independently of whether ZFC has a refutation. In particular, no unspecified recursively enumerable index or consistency-dependent choice occurs in the definition.

1.1 What improves, and what is retained

The exact old degree is

$$126496 \cdot 5^{31} + 27064 = 589042901992797851562527064.$$

The ratio of this degree to the new degree is approximately $2.4799670391 \cdot 10^{18}$. The operation reduction is 660/1442, approximately 45.77%.

These are comparisons on the *same ten-unrestricted-integer branch*. The repository also contains much smaller operation counts with more unknowns or with positivity restrictions, low-degree systems with many witnesses, and nine-positive-variable forms [1]. The present result does not dominate all of these measures. The ten-variable unrestricted-integer reduction itself is Sun's result [6]. Package 31's quintic divisibility merger, coprime square merger, and structural separating weights are also retained, not claimed anew.

The new synthesis changes the input to that finalizer:

$$\begin{aligned} &\text{old: quadratic rows} &&\longrightarrow \text{one quartic sum of squares} &&\longrightarrow \text{positions } 5^j,\\ &\text{new: quadratic rows} &&\longrightarrow \text{Sidon positions} &&\longrightarrow \text{all target digits at once.}\end{aligned}$$

The first row loses the low degree of the system before coding it. The second row preserves it and uses the same packed witness for every equation.

1.2 The mathematical dependency boundary

The proof has three distinct layers. First, the inherited construction relates the literal K215 machine, its marked Wang language, a native-window/Pell compiler, and the positive quadratic parent system. Second, Sections 3 and 4 prove the new simultaneous-digit source reduction. Third, Sections 5 and 6 compose that source with Sun's bridges and the retained finalizer.

The repository provides formal K215 endpoint sources. This article does not run their full proof-assistant dependency tree or infer ZFC provability from a blanket assertion that ZFC proves a proof assistant sound. The mathematical route uses the substantive inherited K215 calibration, formalizable in ZFC. Besides finite simulation and proof enumeration, that calibration includes a seed/quotient model construction, first-order completeness, and the ZF–ZFC refutability equivalence obtained through the constructible universe. The finite programs in this archive verify transcriptions and algebraic claims; they do not prove those entire inherited semantic theorems or establish Con(ZFC).

2 The fixed quadratic parent

We retain the complete coefficient tuple used by package 31. Its four positive integer ports are

$$(\beta, K_0, M_C, M_F),$$

where M_F is the *already shifted* high mask. Appendix A gives the finite recipe; the archive also materializes all four integers. Their respective bit lengths are

$$(1064921,\ 1070004,\ 1064921,\ 1064922).$$

Keeping the whole tuple unchanged avoids any transfer claim between different compiler adapters.

The following parent coordinate names are local to this section:

$$\begin{gathered} J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.\end{gathered}$$

All 31 coordinates are strictly positive. Put

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H_p = 4a + 3, \qquad C_p = q - F - \alpha.$$

The 22 equations, in their original order, are as follows.

Row	Equation
1	$L = q^2 - 1$
2	$N = q(L + 1)$
3	$X = wq$
4	$a = E + sN$
5	$E = X(a - E)$
6	$c = k(a - E) + \eta$
7	$P = E(c - \eta) + k$
8	$\tau^2 = P(P - k) + 1$
9	$D = X + ac + \gamma H_p$
10	$\Delta = a^2 + H_p$
11	$C_2 = c^2$
12	$D^2 = \Delta C_2 + 1$
13	$k = jc - V + 1 + hE$
14	$jc - V = (q - F)(N - q) - C_p L + (M_C + qM_F)J$
15	$(K_0 + w)C_p + q - F - z(q - 1) = 1$
16	$F_2 = f^2$
17	$V = of - c$
18	$g = y^2 - V^2$
19	$I = iC_2$
20	$B_{\text{aux}} = I^2$
21	$B_{\text{aux}} = \Delta(F_2 - 1)$
22	$y^2 - B_{\text{aux}} g = 1$

Every residual is quadratic after expanding the four affine abbreviations. The inherited parent theorem is

$$\exists \mathbf{p} \in \mathbb{Z}_{>0}^{31} \text{ all 22 equations hold} \quad \longleftrightarrow \quad K215 \text{ halts on blank input} \quad \longleftrightarrow \quad \neg\,\mathrm{Con(ZFC)}. \tag{2.1}$$

Its sources are the copied parent proof, coefficient adapter, and K215 endpoint [2, 3, 4].

For each coordinate substitute $p_i = n_i + 1$. Form left side minus right side, and reorder the rows as

$$1, 2, 3, 4, 6, 7, \ldots, 22, 5.$$

Call the resulting polynomials $r_1, \ldots, r_{22} \in \mathbb{Z}[n_1, \ldots, n_{31}]$. The last row satisfies

$$r_{22}(0, \ldots, 0) = 1. \tag{2.2}$$

Thus the all-zero tuple is not a common zero, and the leading coefficient of the later reversed packing will be positive. The successor shift is essential: the parent has positive witnesses, whereas the source coding uses nonnegative digits.

An exact symbolic audit compares the parent circuit with an independently retrieved sparse representation, keeping all four ports as independent symbols. It verifies both the shift and every coefficient used below. Each ordinary coefficient of each normalized row is a polynomial of port

degree at most two, with sum of absolute integer coefficients at most eight. Consequently the completely fixed positive integer

$$H = 8(1 + \beta + K_0 + M_C + M_F)^2 \tag{2.3}$$

bounds the absolute values of all ordinary coefficients of all r_j.

3 Direct quadratic packing at several target positions

3.1 A small Sidon set replaces exponential positions

A finite set is a B_2 set, or Sidon set in the convention used here, if all its unordered pair sums, including repeated summands, are distinct. Use the following 32 positions, in increasing order:

$$\begin{aligned}(w_0, \dots, w_{31}) = (&0, 10, 35, 55, 62, 105, 144, 145,\\ &159, 168, 232, 245, 306, 323, 325, 353,\\ &371, 374, 382, 494, 499, 525, 566, 578,\\ &610, 647, 681, 685, 741, 777, 783, 799).\end{aligned} \tag{3.1}$$

All $32 \cdot 33/2 = 528$ pair sums are distinct modulo 1023. They are therefore distinct as integers as well. Padding a monomial of degree less than two with zero-weight letters makes every monomial of degree at most two correspond to exactly one unordered pair. It follows that

$$\alpha \longmapsto e(\alpha) = \sum_{i=1}^{31} \alpha_i w_i \quad (|\alpha| \le 2)$$

is injective.

The list was obtained from the classical Bose–Chowla construction in $\mathbb{F}_{32^2}$, with an affine change modulo 1023. The generation metadata use the binary polynomial $x^{10} + x^3 + 1$, multiplier 47 and shift 879. The finite pair-sum certificate alone establishes the property required here. Thus no finite-field implementation is a hidden assumption of the particular equation. Bose–Chowla constructions and their affine changes are standard [7, 8]; global priority for this style of coding is not claimed.

Put

$$\begin{aligned} t = 799, \qquad & E_j = 2t + (j-1)(2t+1) \quad (1 \le j \le 22),\\ E_{\max} = 44t + 21 = 35177, \qquad & M = E_{\max} + 2t = 36775.\end{aligned} \tag{3.2}$$

Write $r_j = \sum_\alpha a_{j,\alpha} n^\alpha$, omitting zero terms, and set

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_{i=1}^{31} \alpha_i!\, a_{j,\alpha}, \qquad D_{\mathrm{c}}(Z) = \sum_{j=1}^{22} \sum_{|\alpha| \le 2} c_{j,\alpha} Z^{E_j - e(\alpha)}. \tag{3.3}$$

All exponents are nonnegative. The coefficient records number 136; there are 161 terms when their port polynomials are expanded. The bound $|c_{j,\alpha}| \le 2H$ follows immediately.

Lemma 3.1 (Simultaneous target extraction)**.** *For any* $\mathbf{n} \in \mathbb{N}^{31}$*, put*

$$C_{\mathbf{n}}(Z) = \left(1 + \sum_{i=1}^{31} n_i Z^{w_i}\right)^2.$$

Then, for every $j = 1, \dots, 22$,

$$[Z^{E_j}]\, C_{\mathbf{n}}(Z) D_{\mathrm{c}}(Z) = 2r_j(\mathbf{n}). \tag{3.4}$$

Every coefficient of the product has absolute value at most

$$2H \left(1 + \sum_i n_i\right)^2.$$

Proof. The coefficient of $Z^{e(\alpha)}$ in $C_{\mathbf{n}}$ is exactly

$$\frac{2!}{(2 - |\alpha|)! \prod_i \alpha_i!}\, \mathbf{n}^{\alpha}$$

by injectivity of the padded pair sums. This cancels the factorial multiplier in (3.3).

The reversed support belonging to row j lies in $[E_j - 2t, E_j]$. These intervals are disjoint because successive targets are $2t+1$ apart. After multiplication by $C_{\mathbf{n}}$, row j has support in $[E_j - 2t, E_j + 2t]$, which contains no other target. Thus no other row contributes to the coefficient at E_j.

All coefficients of $C_{\mathbf{n}}$ are nonnegative and their sum is $(1 + \sum_i n_i)^2$. Every coefficient of D_{c} has absolute value at most $2H$, since its row supports are disjoint. The stated product bound follows by convolution. □

The product support intervals may overlap *away from their targets*. No disjointness of those larger intervals is needed. Allowing that overlap is why spacing $2t + 1$, rather than $4t + 1$, is sufficient.

3.2 A small example

Take $w = (0, 1, 3)$, the equations

$$r_1 = n_1 + n_2 - 5, \qquad r_2 = 6 - n_1 n_2,$$

and targets $E_1 = 6, E_2 = 13$. The reversed polynomial is

$$D_{\mathrm{c}}(Z) = 12Z^{13} - Z^9 - 10Z^6 + Z^5 + Z^3.$$

Multiplication by $(1 + n_1 Z + n_2 Z^3)^2$ produces

$$[Z^6] C_{\mathbf{n}} D_{\mathrm{c}} = 2(n_1 + n_2 - 5), \qquad [Z^{13}] C_{\mathbf{n}} D_{\mathrm{c}} = 2(6 - n_1 n_2).$$

The generic support enclosures $[0, 12]$ and $[7, 19]$ overlap. Nevertheless, each tested coefficient belongs to just one row. This example is also the independent positive and negative control used in `code/check_multitarget_toy.py`.

3.3 The structural gain for arbitrary quadratic systems

For n variables and r quadratic rows, use a B_2 set of $n + 1$ positions including zero. Bose–Chowla sets give maximum position $t = O(n^2)$: take a power of two $q \geq n + 1$ with $q < 2(n + 1)$, choose $n + 1$ members of the q-element construction modulo $q^2 - 1$, and translate one to zero. The same argument uses

$$E_j = 2t + (j - 1)(2t + 1), \qquad M = (2r + 2)t + r - 1 = O(rn^2).$$

This is a polynomial bound on digit-position span. It replaces the exponential span of the old general-purpose monomial coding for this class of systems. It is a statement about this coding step, not a complexity bound for solving the equations.

4 The explicit source and its two directions

4.1 A fixed coefficient bound

Set once and for all

$$\beta_S = 2^{2140030}. \tag{4.1}$$

This is an even power of two and satisfies

$$\beta_S > 4 \cdot 33^2 H. \tag{4.2}$$

For a transparent bound, each native port is less than $2^{1070004}$. Thus $1+\beta+K_0+M_C+M_F < 2^{1070007}$, $H < 2^{2140017}$, and $4 \cdot 33^2 < 2^{13}$. The exact integer check is stronger: H has 2,140,010 bits and $4 \cdot 33^2 H$ has 2,140,022 bits. The complete bytes of β_S are supplied, rather than just its residue modulo a few primes.

4.2 Two-variable polynomial source

From this point forward the letters refer to the finalizer, not the positive parent coordinates. Start with integer unknowns f, g and define

$$\begin{aligned} b &= 1 + 3f, \qquad \mathcal{B} = \beta_S b^2, \qquad \ell = \sum_{i=1}^{31} \mathcal{B}^{w_i}, \\ N_0 &= \mathcal{B}^{t+1}, \qquad N_1 = 4\mathcal{B}^{M+1}, \quad N = N_0 N_1. \end{aligned} \tag{4.3}$$

Define the masks and packed words by

$$\begin{aligned} M_{\mathrm{c}} &= N_0 - 1 - (b-1)\ell, \\ Q_{\mathrm{c}} &= (\mathcal{B}-2) \sum_{j=1}^{22} \mathcal{B}^{E_j}, \\ T_{\mathrm{c}} &= M_{\mathrm{c}} + Q_{\mathrm{c}} N_0, \\ J_{\mathrm{c}} &= 2(1+g)^2 D_{\mathrm{c}}(\mathcal{B}) + \sum_{i=0}^{M} \mathcal{B}^{i+1}, \\ S_{\mathrm{c}} &= g + J_{\mathrm{c}} N_0, \\ R_{\mathrm{c}} &= (S_{\mathrm{c}} + T_{\mathrm{c}} + 1)N + T_{\mathrm{c}} + 1. \end{aligned} \tag{4.4}$$

Finally put

$$X = (N-1)R_{\mathrm{c}}, \qquad Y = N^2, \qquad \mathscr{C} = b\mathcal{B}^t. \tag{4.5}$$

In particular,

$$N = 4\mathcal{B}^{37576}, \qquad \mathscr{C} = \beta_S^{799} b^{1599} = 2^{1709883970} b^{1599}.$$

Every expression in (4.3)–(4.5) is an integer polynomial in f, g.

Proposition 4.1 (Source interface)**.** *The following statements hold.*

(i) *If the r_j have a common zero in $\mathbb{N}^{31}$, then for every sufficiently large positive integer f for which b is both a square and a power of two, there is an integer g such that*

$$b \le g < \mathscr{C}, \qquad Y \mid \binom{2X}{X}.$$

(ii) *If $f \neq 0$, b is a square and a power of two,*

$$0 \le g < 2\mathscr{C}, \qquad Y \mid \binom{2X}{X},$$

then the r_j have a common zero in $\mathbb{N}^{31}$.

(iii) *Already under $f \neq 0$, b a square, and $0 \le g < 2\mathscr{C}$, without assuming that b is a power of two, one has*

$$0 \le S_{\mathrm{c}}, T_{\mathrm{c}} < N, \qquad 3 \mid X, \qquad X > 3b, \qquad Y > \max\{b, 256\}.$$

4.3 Ranges before recovering a power-of-two base

Suppose $f \neq 0$ and b is a square. As $b = 1 + 3f$, this forces $f > 0$, $b \ge 4$, and $\mathcal{B} > 2b$. The highest coefficient of D_{c}, at $E_{\max}$, is $2r_{22}(0) = 2$. All remaining coefficients have absolute value at most $2H$, so

$$D_{\mathrm{c}}(\mathcal{B}) \ge 2\mathcal{B}^{E_{\max}} - 2H\frac{\mathcal{B}^{E_{\max}} - 1}{\mathcal{B} - 1} > 0.$$

The digits of M_{c} are $\mathcal{B}-b$ at the selected positions and $\mathcal{B}-1$ elsewhere below N_0. Thus $0 \le M_{\mathrm{c}} < N_0$. Similarly $0 \le Q_{\mathrm{c}} < \mathcal{B}^{E_{\max}+1} < N_1$, and hence $0 \le T_{\mathrm{c}} < N$.

For an arbitrary integer $0 \le g < 2b\mathcal{B}^t$, integrality gives

$$1 + g \le 2b\mathcal{B}^t, \qquad g < N_0.$$

The coefficient bound implies $16Hb^2 \le \mathcal{B} - 2$. Therefore

$$\begin{aligned} 0 < J_{\mathrm{c}} &\le 16Hb^2\mathcal{B}^{2t}\sum_{i=0}^{E_{\max}} \mathcal{B}^i + \sum_{i=0}^{M} \mathcal{B}^{i+1} \\ &< \frac{16Hb^2 + \mathcal{B}}{\mathcal{B} - 1}\mathcal{B}^{M+1} \le 2\mathcal{B}^{M+1} < N_1. \end{aligned}$$

It follows that $0 \le S_{\mathrm{c}} < N$. This argument works for every integer in the broad permitted interval, before any digit recovery.

Since $\beta_S \equiv b \equiv 1 \pmod 3$, we have $\mathcal{B} \equiv 1 \pmod 3$ and $N = 4\mathcal{B}^{37576} \equiv 1 \pmod 3$. Thus $3 \mid X$. Further, $R_{\mathrm{c}} \ge N + 1 > \mathcal{B} > b$ and $N \ge 4$, so $X > 3b$. The displayed fixed base also gives $Y = N^2 > \max\{b, 256\}$. This proves part (iii).

4.4 Recovering the digits and testing every row

Assume additionally that b is a power of two. Then $\mathcal{B}, N_0, N$ are powers of two. Let $\tau_2(u, v)$ be the number of binary carries when adding nonnegative integers u, v. Equivalently,

$$\tau_2(u, v) = \mathrm{pop}(u) + \mathrm{pop}(v) - \mathrm{pop}(u + v),$$

where pop denotes the number of 1-bits. In particular, $\tau_2(u, v) = 0$ means their 1-bits are disjoint. For $0 \le g < 2\mathscr{C} < N_0$, the low mask gives

$$\tau_2(g, M_{\mathrm{c}}) = 0 \quad \Longleftrightarrow \quad g = \sum_{i=1}^{31} n_i\mathcal{B}^{w_i}, \qquad 0 \le n_i < b. \tag{4.6}$$

An unselected block is masked by $\mathcal{B}-1$, forcing its digit to zero. A selected block is masked by $\mathcal{B}-b$; its available low bits represent exactly the integers below b.

For a word of this form, define

$$K_{\mathrm{c}} = (1+g)^2 D_{\mathrm{c}}(\mathcal{B}) + \frac{\mathcal{B}}{2}\sum_{i=0}^{M}\mathcal{B}^i, \qquad J_{\mathrm{c}} = 2K_{\mathrm{c}}.$$

This expression is used only in the proof; it is not a division gate in the polynomial circuit. By Lemma 3.1,

$$\left|[Z^i]C_{\mathbf{n}}D_{\mathrm{c}}\right| \le 2H(1+\textstyle\sum_i n_i)^2 < \mathcal{B}/2.$$

Hence the base-$\mathcal{B}$ digits of K_{c} are exactly

$$\mathcal{B}/2 + [Z^i]C_{\mathbf{n}}(Z)D_{\mathrm{c}}(Z), \quad 0 \le i \le M,$$

all strictly between zero and $\mathcal{B}$. In particular, its digit at E_j is $\mathcal{B}/2 + 2r_j(\mathbf{n})$. For $1 \le e < \mathcal{B}$, the no-carry condition

$$\tau_2(e, \mathcal{B}/2-1) = 0$$

holds exactly when $e = \mathcal{B}/2$. Applying that test at every target, and then shifting both summands left by one bit, gives

$$\tau_2(J_{\mathrm{c}}, Q_{\mathrm{c}}) = 0 \quad\Longleftrightarrow\quad r_1(\mathbf{n}) = \cdots = r_{22}(\mathbf{n}) = 0. \tag{4.7}$$

The lower and upper masks occupy separate blocks, so

$$\tau_2(S_{\mathrm{c}}, T_{\mathrm{c}}) = 0 \quad\Longleftrightarrow\quad \tau_2(g, M_{\mathrm{c}}) = \tau_2(J_{\mathrm{c}}, Q_{\mathrm{c}}) = 0.$$

4.5 An elementary central-binomial bridge

Lemma 4.2. *Let $N = 2^e$ and $0 \le S, T < N$. Put*

$$X = (N-1)\big((S+T+1)N + T + 1\big).$$

Then

$$N^2 \mid \binom{2X}{X} \quad\Longleftrightarrow\quad \tau_2(S,T) = 0.$$

Proof. The exact base-N decomposition is

$$X = (S+T)N^2 + (N-1-S)N + (N-1-T).$$

The two low blocks are e-bit complements. It follows that

$$\mathrm{pop}(X) = 2e + \mathrm{pop}(S+T) - \mathrm{pop}(S) - \mathrm{pop}(T) = 2e - \tau_2(S,T).$$

Legendre's formula $\nu_2(n!) = n - \mathrm{pop}(n)$ gives $\nu_2\binom{2X}{X} = \mathrm{pop}(X)$. This valuation is at least $2e$ exactly when the nonnegative carry count is zero. □

This is the binary specialization of the carry construction used by Sun [5]; the proof above supplies the exact identity needed here. Combining the lemma with (4.6) and (4.7) proves the reverse direction of Proposition 4.1.

For completeness, start with a common zero $\mathbf{n}$. Choose $b = 2^{2q} > \max_i n_i$, as large as desired, and put $f = (b-1)/3$. This is a positive integer. Set $g = \sum_i n_i \mathcal{B}^{w_i}$. The tuple is nonzero by (2.2); hence $g \ge \mathcal{B} \ge b$. The geometric-series bound and $\mathcal{B} > b$ give $g < b\mathcal{B}^t = \mathscr{C}$. Both masks pass, so Lemma 4.2 yields the required binomial divisibility. This proves part (i) and completes the source proof.

5 The complete ten-integer polynomial

The source just proved replaces the source of package 31. All subsequent polynomials are retained from its structural three-square finalizer [2], whose Lucas ingredients come from [5, 6]. Introduce the remaining eight integer unknowns h, k, l, w, x, y, z, m. The complete list of unknowns is now exactly

$$f, g, h, k, l, w, x, y, z, m.$$

Starting with $b, X, Y, \mathscr{C}$ from Section 4, define

$$\begin{aligned}
&L = lY, && U = 2LX, && V = 4gwY, && W = bw,\\
&A = U(V+1), && B_1 = 2X+1, && C = B_1 + (A-2)h,\\
&P_n = U^2V, && K = X + 1 + k(P_n - 2),\\
&S = 2A - 5, && T = 3WC - 2(W^2-1),\\
&R_2 = (P_n^2 - 4)K^2 + 4.
\end{aligned} \tag{5.1}$$

Put

$$\begin{aligned}
\Delta_b &= 8\mathscr{C}^3 gK^2 - g^2\big(32(C - KL)^2\mathscr{C}^3 + g^2K^2\big),\\
O &= f^2l^2\Delta_b, \qquad \xi = zx.
\end{aligned} \tag{5.2}$$

The modified Lucas certificate is

$$\begin{aligned}
&D = (A^2-4)C^2 + 4, && E = C^2D\xi,\\
&F = 4(A^2-4)E^2 + 1, && G^* = 2(A+2)(A-2)^2E^2 - 1,\\
&\Omega = 16E, && H^* = C + (B_1 - C)F + \Omega Fy,\\
&I^* = ((G^*)^2 - 1)(H^*)^2 + 1.
\end{aligned} \tag{5.3}$$

The divisibility merger and radicands are

$$\begin{aligned}
&\Sigma = \xi S^5, && r = T + Sy,\\
&\Theta = (r^3 - rS^2 - S^3)(r^2 + 23S^2),\\
&r_0 = b(1 + (3O-4)b^2z^2), && r_1 = R_2, && r_2 = DFI^*,\\
&v_1 = r_0, && v_2 = r_0(P_n^2 + K^2 + 3).
\end{aligned} \tag{5.4}$$

Finally set

$$\begin{aligned}
Z_{\mathrm{n}} &= \Sigma m - \Theta,\\
a_0 &= \Sigma^2 r_0, && a_1 = \Sigma^2 v_1^2 r_1, \quad a_2 = \Sigma^2 v_2^2 r_2,\\
V_{\mathrm{n}} &= Z_{\mathrm{n}}^2 + a_2 - a_0 - a_1,\\
W_{\mathrm{n}} &= V_{\mathrm{n}}^2 + 4Z_{\mathrm{n}}^2 a_2 - 4a_0a_1.
\end{aligned} \tag{5.5}$$

The promised equation is the following single equation:

$$\boxed{F_{10} = W_{\mathrm{n}}^2 - 16V_{\mathrm{n}}^2 Z_{\mathrm{n}}^2 a_2 = 0.} \tag{5.6}$$

Equations (5.1)–(5.5) are substitution rules, not additional equations with additional unknowns. In particular, (5.6) contains no radicals, variable exponents, divisibility predicates, order conditions, or hidden parameters.

6 Why the finalizer preserves the equivalence

6.1 The inherited bridge in the interface form used here

Let $u_n(A)$ be the Lucas sequence $u_0 = 0, u_1 = 1$, $u_{n+1} = Au_n - u_{n-1}$, with its usual extension to integer indices. The formulas above are those of Sun's modified binomial bridge, followed by the package-31 changes.

The literal statements of Sun's 2026 Propositions 6.1 and 6.2 refer to his earlier source recipe. We use their *proofs parametrically.* In those proofs the source is used only through: its forward and reverse binomial-divisibility implications; $b = 1 + 3f$ (the parameter $a = 0$); the fixed form $\mathscr{C} = 2^c b^d$ with $c = 1709883970$ even and $d = 1599$ odd; and the size and congruence conditions in Proposition 4.1. All these properties were proved above. No appeal to the old quartic coding identity is left in the composition.

In this form the inherited bridge has the following consequences:

1. From a common nonnegative parent zero, there are f, g, h, k, l, w with $A > 2$ even, $B_1 > 0$ odd,
$$b \in \mathrm{Sq}(\mathbb{Z}), \quad b \le g < \mathscr{C}, \quad C = u_{B_1}(A), \quad R_2 \in \mathrm{Sq}(\mathbb{Z}), \quad S \mid T, \quad O > 0.$$

2. Conversely, for the displayed formulas, the conditions
$$b \in \mathrm{Sq}(\mathbb{Z}), \quad O > 0, \quad \xi \neq 0, \quad DFI^* \in \mathrm{Sq}(\mathbb{Z}), \quad R_2 \in \mathrm{Sq}(\mathbb{Z}), \quad S \mid T$$
imply a common nonnegative parent zero.

3. If $A > 2$ is even, $B_1 > 0$ is odd, and $C = u_{B_1}(A)$, then for every positive integer d there is $\xi > 0$ divisible by d such that *each* residue class modulo ξ contains a y making DFI^* a square.

The last assertion is the full residue-class interpolation theorem [6, Theorem 4.1]; the first two use its Propositions 6.1–6.2 and the cited Sun-2021 Lucas estimates.

For clarity, the reverse proof first obtains $f, l \neq 0$, $g > 0$, $K \neq 0$, and
$$0 < g < 2\mathscr{C}, \qquad 4(C - KL)^2 < K^2$$
from $O > 0$. The already established source ranges then apply. The opening argument of Sun's 2021 Theorem 4.13 gives $W \neq 0$ and $1 < B_1 < |A|/2 - 1$. Also A is even, B_1 is odd, and $A - 2 \mid C - B_1$ by definition. These are the applicability conditions for the reverse modified Lucas certificate. It recovers $C = u_{B_1}(A)$, and the remaining Lucas and approximation arguments give
$$K = u_{X+1}(P_n), \qquad W = 2^{B_1}, \qquad Y \mid \binom{2X}{X}.$$
As $W = bw$ and $b > 0$, b is a power of two. Proposition 4.1(ii) finishes that implication. This ordering is why the source bounds for arbitrary permitted g were needed before power-of-two recovery.

6.2 The quintic divisibility merger

The package-31 merger uses
$$\pi(v) = (v^3 - v - 1)(v^2 + 23).$$

It has no rational zero but has a root modulo every positive integer. Here is the local argument. The cubic has discriminant -23. For an odd prime $p \neq 23$, a root of the cubic is simple. If it has no root, it is irreducible of degree three, so Frobenius acts as an even 3-cycle and its discriminant is a square modulo p. Then the quadratic has a simple root. At $p = 23$, the cubic has the simple root 3. These roots lift to every odd prime power. At 2, $-23 \equiv 1 \pmod 8$ is a square in the 2-adic integers. The Chinese remainder theorem combines the prime-power roots. The rational-root test excludes a rational root of the cubic, and the quadratic is positive on $\mathbb{Q}$.

The integer $S = 2A - 5$ is always odd and nonzero. If $S \mid T$, write $T = Sq$. For any $\xi \neq 0$, choose a root c of π modulo $|\xi|$ and prescribe

$$y \equiv c - q \pmod{\xi}.$$

Then $\Theta = S^5\pi(q + y)$, so $\Sigma = \xi S^5 \mid \Theta$. Conversely, if $S^5 \mid \Theta$ but $S \nmid T$, choose a prime with $\alpha = \nu_p(S) > \beta = \nu_p(T)$. The two homogeneous factors of Θ have valuations 3β and 2β; their leading terms have uniquely least valuation. Thus $\nu_p(\Theta) = 5\beta < 5\alpha$, a contradiction. Furthermore $\Theta \neq 0$, because its vanishing would give a rational zero $(T + Sy)/S$ of π.

6.3 The coprime square merger

Suppose $z \neq 0$ and r_0 is a square. The factors b and $1 + (3O - 4)b^2z^2$ are coprime. The second factor cannot vanish: vanishing forces $|b| = |z| = 1$, while $b \equiv 1 \pmod 3$ then gives $b = 1$, $f = 0$, and $O = 0$, contradicting the vanishing equation. Unique factorization therefore makes b a signed square. Since a negative nonzero square cannot be 1 modulo 3, b is a positive square and the second factor is also a square. If $O \leq 0$, that factor is at most $1 - 4(bz)^2 \leq -3$, impossible. Thus $O > 0$.

For the converse, use positive bridge witnesses. The positive integer Δ_b is a multiple of g, so

$$O = f^2l^2\Delta_b \geq g \geq b \geq 4.$$

The integer $d = 3O - 4$ is positive and is 2 modulo 3, hence nonsquare. A positive Pell unit for d, raised to a suitable finite order modulo b, gives

$$u^2 - dv^2 = 1, \qquad v > 0, \qquad b \mid v.$$

Set $z = v/b > 0$. The bracket in r_0 is u^2, so r_0 is a square. Interpolation may now choose a positive ξ divisible by this z, followed by a y in the required residue class. The order of these choices is legitimate; no residue class is prescribed before its modulus exists.

6.4 Separating three square roots

For every integer assignment with $z \neq 0$, one has $|r_0| > 1$. Indeed, $r_0 = 0, \pm 1$ would force $|b| = 1$; then $b = 1, f = O = 0$, but $r_0 = 1 - 4z^2 \leq -3$. Also

$$\sqrt{|r_1|} \leq P_n^2 + K^2 + 2.$$

For example, square the right side and subtract $(P_n^2 + 4)K^2 + 4$, an upper bound on $|r_1|$; the remainder is

$$P_n^4 + K^4 + P_n^2K^2 + 4P_n^2 \geq 0.$$

It follows that

$$|v_1| > \sqrt{|r_0|}, \qquad |v_2| > \sqrt{|r_0|} + |v_1|\sqrt{|r_1|}. \tag{6.1}$$

These inequalities imply that a rational number of the form

$$\varepsilon_0\sqrt{r_0}+\varepsilon_1 v_1\sqrt{r_1}+\varepsilon_2 v_2\sqrt{r_2},\qquad \varepsilon_i\in\{1,-1\},$$

can occur only when all three radicands are integer squares. To see this, work in the multiquadratic extension generated by the roots. If a root is irrational, an automorphism changes it; take the largest-index changed nonzero term. Its change has absolute value at least twice its weight, since a nonzero integral radicand has root absolute value at least one. The sum of the earlier changes has strictly smaller absolute value by (6.1). Thus the entire sum changes and is not rational. The same argument covers negative radicands, using complex absolute value, and does not require independent square classes. An integral radicand with rational square root is a nonnegative integer square.

6.5 The norm and both implications

As an exact polynomial identity in $Z_{\mathrm{n}}, a_0, a_1, a_2$, the right side of (5.6) is

$$\prod_{\varepsilon\in\{1,-1\}^3}\left(Z_{\mathrm{n}}+\varepsilon_0\sqrt{a_0}+\varepsilon_1\sqrt{a_1}+\varepsilon_2\sqrt{a_2}\right). \tag{6.2}$$

The radicals cancel. Repeatedly multiplying conjugate factors gives exactly $W_{\mathrm{n}}^2-16V_{\mathrm{n}}^2Z_{\mathrm{n}}^2a_2$; the verifier also expands and compares both 35-term integer polynomials symbolically.

Given a common nonnegative parent zero, take positive bridge witnesses. Choose $z>0$ by the Pell step, $\xi>0$ divisible by z by interpolation, and put $x=\xi/z$. Prescribe y using the quintic local root and then use interpolation in that residue class. All r_i are squares, $\Sigma\neq 0$, and $\Sigma\mid\Theta$. Choose m so that

$$\Sigma m-\Theta=-\Sigma(\varepsilon_0\sqrt{r_0}+\varepsilon_1 v_1\sqrt{r_1}+\varepsilon_2 v_2\sqrt{r_2})$$

for any selection of signs. This is an integer choice. One factor of (6.2) vanishes, giving $F_{10}=0$.

Conversely suppose $F_{10}=0$. We always have $S\neq 0$ and $\Theta\neq 0$. If $\Sigma=0$, all $a_i=0$ and the polynomial reduces to $\Theta^8\neq 0$, a contradiction. Therefore $\xi=zx\neq 0$ and $z\neq 0$. Dividing a vanishing norm factor by Σ makes a weighted signed sum of the roots of the r_i rational. Separation forces all three radicands to be squares, and the sum is then an integer. Hence $\Sigma\mid\Theta$. The quintic merger gives $S\mid T$, and the coprime square merger gives b square and $O>0$. The reverse bridge now supplies a common nonnegative parent zero. Combining with (2.1) proves (1.1). Classical negation gives the assertion about absence of solutions.

7 Exact degree and the 782-operation circuit

7.1 Evaluation without expanding the enormous polynomial

The literal polynomial is specified by `data/polynomial.json`, a finite directed acyclic graph whose non-leaf gates are binary addition, subtraction, and multiplication. The fixed nonliteral leaves have exact finite recipes in `data/coefficient_recipes.json`. There are 16 such leaves, including β_S; the ordinary small integer leaves are literal. The four native ports are not unknowns.

The builder evaluates D_{c} efficiently by homogenizing each quadratic row. Put

$$U_0=\mathcal{B}^t,\qquad U_i=\mathcal{B}^{t-w_i},\qquad Q_j=\sum_{\alpha}c_{j,\alpha}U_0^{2-|\alpha|}\prod_i U_i^{\alpha_i}.$$

Then the exact identity

$$D_{\mathrm{c}}(\mathcal{B}) = \sum_{j=1}^{22} \mathcal{B}^{(j-1)(2t+1)} Q_j$$

allows shared multivariate Horner evaluation of the row forms and ordinary Horner evaluation across rows. It preserves the literal coefficient-extraction polynomial.

Fixed powers use shared repeated squaring. For a geometric sum $G_n = 1 + \mathcal{B} + \cdots + \mathcal{B}^{n-1}$, the builder uses

$$G_{2n} = G_n(1 + \mathcal{B}^n), \qquad G_{2n+1} = G_{2n} + \mathcal{B}^{2n}.$$

Thus the large sum limits do not entail one gate per summand. Every fixed power is expanded into binary gates before counting.

The operation convention matches package 31: each binary addition, subtraction, or multiplication costs one; fixed leaves, arithmetic involving only fixed constants, and reuse are free. Multiplication of a variable-dependent expression by a fixed numeral is charged. Under this convention the live graph has

$$221 + 25 + 536 = 782$$

operations. This is a certified count of the supplied evaluator, not an arithmetic-circuit lower bound.

7.2 A symbolic proof of exact degree

Let

$$u = M + t + 2 = 47t + 23 = 37576.$$

Every fixed coefficient has degree zero and every unknown has degree one. The highest term of $D_{\mathrm{c}}(\mathcal{B})$ is nonzero because the last normalized row has constant coefficient one. The geometric sum in J_{c} has degree $2(M+1)$, strictly greater than the other summand's degree. It follows successively that

expression	N	S_{c}	T_{c}	X	Y	$\mathscr{C}$
degree	$2u$	$2u$	$2u-4t$	$6u$	$4u$	$2t+1.$

In particular,

$$\deg X = 225456, \qquad \deg Y = 150304, \qquad \deg \mathscr{C} = 1599.$$

Write $d_X = \deg X$, $d_Y = \deg Y$, and $d_C = \deg \mathscr{C}$. Direct propagation through the retained finalizer gives

$$\begin{aligned}
\deg Z_{\mathrm{n}} &= 5d_X + 10d_Y + 30,\\
\deg a_0 &= 14d_X + 28d_Y + 3d_C + 57,\\
\deg a_1 &= 26d_X + 48d_Y + 6d_C + 98,\\
\deg a_2 &= 114d_X + 224d_Y + 6d_C + 440.
\end{aligned}$$

The last degree strictly exceeds the degrees of $a_0, a_1, Z_{\mathrm{n}}^2$. Consequently the highest homogeneous part of F_{10} is the fourth power of the highest homogeneous part of a_2, which is nonzero. Hence

$$\begin{aligned}
\deg F_{10} &= 456d_X + 896d_Y + 24d_C + 1760\\
&= 6320u + 48t + 1784\\
&= 297088t + 147144 = 237520456.
\end{aligned} \tag{7.1}$$

Appendix B records the intermediate degrees and the strict-dominance argument for the few expressions with subtraction.

There is also an independent exact lower-bound certificate. After substituting all ten variables by one indeterminate T, the coefficient at the circuit's total-degree upper bound is

$$[T^{237520456}]F_{10}(T,\dots,T) \equiv 449536588 \pmod{1000000007}.$$

This nonzero residue proves attainment of the upper bound. It is a coefficient computation, not random sampling or numerical interpolation.

7.3 Coefficient size is a separate measure

The new source constant has 2,140,031 bits, whereas package 31's old source constant $2^{4269882}$ has 4,269,883 bits. The four native compiler ports are exactly unchanged. These numbers describe fixed leaves in the circuit representation. They are *not* the heights of coefficients after fully expanding F_{10}. Those expanded heights have not been measured, and no claim about their optimality is made.

8 Verification, reproducibility, and proof status

From the archive root, run

```
python3 code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The mathematical build and audits use only the Python standard library. No network access or proof search is required. Run Python without `-O`, since assertions are part of the checking process.

The verifier reconstructs the coefficient payloads, regenerates the polynomial, and checks the following distinct claims.

Check	What the result establishes
Native tuple binding	All four raw payload hashes equal package 31's pinned tuple; 430 literal machine actions agree with the transition JSON.
Native finite geometry	69,230 exact scalar identities from the inherited coefficient builder are rechecked. They certify its finite geometry, not the entire unbounded compiler theorem.
Parent and target algebra	Exact sparse polynomial comparison with independent parent data; all 528 pair sums; all 22 target-coefficient identities; normalized row and coefficient bounds.
Independent circuit evaluator	No imports from the builder; 34,848 modular coefficient diagnostics and 6,720 named-node comparisons in 120 complete circuit cases. These are transcription checks, not universal polynomial identity proofs.
Exact norm identity	All eight signed factors expand to the stated 35-term radical-free integer polynomial.

Check	What the result establishes
Exact degree	Structural leading-term proof plus nonzero diagonal leading coefficient modulo a prime.
Small source controls	672 exact cases, four accepting packed tuples across the two chosen bases, and 400 rejected malformed words. The tests use a small soluble two-equation system.

All checks completed successfully. The machine is not run until it halts, and none of these programs decides whether a ZFC refutation exists. They are practical evidence against coding or transcription errors around a mathematical reduction with an unbounded existential meaning.

The core files are:

- `report.tex`, `report.pdf`: this proof and its rendered form;
- `data/polynomial.json`: the literal ten-variable equation;
- `data/weights.json`, `data/coefficient_recipes.json`: the finite source data and exact fixed-leaf recipes;
- `coefficients/`: complete native tuple, builder, proof adapter, literal tile and transition tables, and upstream notices;
- `code/`: builder, exact algebraic checks, independent evaluator, and one-command verification;
- `certificates/`: results, hashes, operation and degree receipts;
- `upstream/`: the pinned parent data, inherited finalizer proof, K215 endpoint and literal machine, and focused mathematical audit notes.

The main result uses no artifact from `research_candidate/`. That directory has a different, explicitly incomplete role described next.

9 Further reductions worth pursuing

9.1 Reduce the weight span or exploit the actual supports

For this finalizer and 22 rows, equation (7.1) shows that lowering t by one lowers the degree by exactly 297,088. The present $t = 799$ is a verified construction, not an optimum 32-mark ruler. A shorter certified B_2 set immediately improves the same proof and need not change any inherited theorem.

More substantially, a full B_2 set separates *all* quadratic monomials, while only 136 weighted records occur in this parent. A support-aware construction might permit some collisions provided they cannot contaminate any of the 22 required targets. Such a proposal must replace the full target-extraction proof, rather than merely demonstrate several successful evaluations. Jointly choosing weights, row order, and nonuniform target gaps is a concrete finite optimization problem with exact symbolic certificates available.

9.2 Reuse the new source on other finalizers

The generic source proposition is separable from the Sun-2026 ten-integer route. Rebuilding the published Sun-2021 route or the repository's other low-variable finalizers with a direct system source may reduce their degrees substantially. These are proposed applications, not additional certified results of this archive. Likewise, eliminating another positive parent witness would require a complete sign and reconstruction argument; no such elimination is claimed here.

9.3 A smaller native coefficient candidate, with an open obligation

A parallel line of work factors the receiver part of the tile compiler. It replaces 488 receiver selectors by two mode selectors and 17 Boolean payload selectors. The candidate has 457 selectors in total and admits materialized coefficient payloads with radix exponent 143 and cell length 4829. Its four raw payloads total 346,098 bytes, compared with 506,664 bytes in package 36, a 31.69% decrease.

Those numbers describe a candidate, not an established replacement tuple. The local decoder and finite convolution checks pass, but a new empty cell may contain nonzero payload bits while its occupancy bits vanish. The inherited binary-alignment and upper-dummy argument must be audited or extended to cover those patterns. Without that unbounded soundness step, the candidate cannot be substituted into the consistency equation. The exact outstanding obligation, proposed decoder, code, notices, and finite receipts are retained in `research_candidate/FACTORIZED_RECEIVERS.md`.

Keeping this obligation visible is useful: a smaller finite coefficient file alone does not establish the compiler's semantic equivalence. The main construction avoids that issue by using the unchanged, already documented package-31 tuple.

9.4 Formalize the new interface

The natural next proof-assistant target is the generic simultaneous-target lemma and source proposition, with a finite B_2 certificate as input. The exact parent polynomials and norm identity then become finite reflection checks. A full end-to-end formalization would additionally need the native compiler and the Lucas/interpolation bridges, whose ordinary mathematical statements are still explicit dependencies.

The present result is a substantial structural and quantitative improvement over the reviewed ten-integer construction. It does not prove the shortest possible equation, a lower bound on unknowns, or worldwide priority for digit coding of systems. Classical Sidon constructions, Sun's coding methods, and prior work on existential arithmetization of systems remain relevant antecedents [7, 8, 5, 9].

A The four fixed native coefficient ports

This appendix restates the literal arithmetic recipe of the retained package-31 coefficient adapter [2]. It is included to make clear that the coefficients are finite specified integers, not parameters whose existence depends on an undecided statement.

Put

$$\rho = 2^{121}, \qquad L_{\mathrm{cell}} = 8801, \qquad d = 1064921, \qquad B_{\mathrm{native}} = 2^{d}.$$

The ordered 943 tiles are the literal quadruples $(x_{i0}, x_{i1}, x_{i2}, x_{i3})$, in west, north, east, south order, in `coefficients/upstream/wang943_tiles.json`. The included builder reconstructs this list from

the 430 literal K215 actions in `coefficients/upstream/k215_transitions.json`. Selector 0 is the start tile, selector 1 the impossible tile, and the remaining 941 tiles are lexicographically ordered.

Define the 989-element support

$$\mathcal{E} = \{0,\dots,942\} \cup \{983,\dots,992\} \cup \{1003,\dots,1032\} \\ \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.$$

Let $W_j = 2^{10+2j}$ for $0 \le j < 44$. For a tile selector i, set

$$c_i = 1 + \sum_{t=0}^{3}\sum_{j=0}^{9} \mathrm{bit}_j(x_{it}+512)\, W_{10t+j} + W_{40} + W_{41} + W_{42} + W_{43}.$$

At copy positions $\mathrm{offset}_t + j$, for

$$(\mathrm{offset}_0,\dots,\mathrm{offset}_3) = (1023, 1013, 1003, 983),$$

put $c_{\mathrm{offset}_t+j} = W_{10t+j}$. The four anchor coefficients are

$$(c_{1095}, c_{1096}, c_{2195}, c_{3296}) = (W_{40}, W_{41}, W_{42}, W_{43}).$$

Finally put $c_{1063} = c_{1064} = 0$.

Define

$$\mu = 1022 + \sum_{j=0}^{43} W_j, \qquad Z_p = (2^{119}-1) \sum_{e=8792}^{8798} \rho^e + (2^{27}-1)\rho^{8799},$$

and

$$D_C = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2200} + \sum_{e\in\mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}),$$

$$D_R = \rho^{42},$$

$$M_C = B_{\mathrm{native}} - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\neq 1}} \rho^e - 2\rho^{1063},$$

$$M_F^{\mathrm{native}} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9}(\rho^{1013+j} + \rho^{1065+j}) + \rho^{2195} + \rho^{3296} + Z_p.$$

The four ports, as one indivisible tuple, are

$$\boxed{\beta = B_{\mathrm{native}} - 1, \qquad K_0 = D_C + B_{\mathrm{native}} D_R, \qquad M_C, \qquad M_F = M_F^{\mathrm{native}} + B_{\mathrm{native}} - 1.}$$

The materialized payloads are unsigned little-endian integers; the `.bin.gz` files decompress to those exact bytes. The coefficient builder and binding manifests specify their hashes. The literal machine file has SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

The newly introduced source constant β_S, also unsigned little-endian, has SHA-256

`b5505995484d5993ce1c0bba8f6fec9d8a38e2d8c5b7b2ac535557bcf48dfdfe.`

Checksums bind files; the arithmetic recipe and inherited compiler proof give their mathematical meaning.

B Detailed degree ledger

Let $x = d_X$, $y = d_Y$, $c = d_C$ in this appendix only. These letters denote degrees, not the polynomial unknowns.

Expression	Exact degree
L	$y + 1$
U	$x + y + 1$
V	$y + 2$
W	2
A	$x + 2y + 3$
B_1	x
C	$x + 2y + 4$
P_n	$2x + 3y + 4$
K	$2x + 3y + 5$
S	$x + 2y + 3$
T	$x + 2y + 6$
R_2	$8x + 12y + 18$
Δ_b	$4x + 8y + 3c + 14$
O	$4x + 8y + 3c + 18$
r_0	$4x + 8y + 3c + 23$
D	$4x + 8y + 14$
E	$6x + 12y + 24$
F	$14x + 28y + 54$
G^*	$15x + 30y + 57$
Ω	$6x + 12y + 24$
H^*	$20x + 40y + 79$
I^*	$70x + 140y + 272$
r_1	$8x + 12y + 18$
r_2	$88x + 176y + 340$
Σ	$5x + 10y + 17$
Θ	$5x + 10y + 30$
Z_{n}	$5x + 10y + 30$
a_0	$14x + 28y + 3c + 57$
a_1	$26x + 48y + 6c + 98$
a_2	$114x + 224y + 6c + 440$

In Δ_b, the term of greatest degree is $-32g^2\mathscr{C}^3K^2L^2$; its degree strictly exceeds the other terms. In $r = T + Sy$, the greatest-degree term comes from T, so the greatest-degree term of Θ is the fifth power of that term. The term Θ strictly dominates Σm. Finally a_2 strictly dominates $a_0, a_1, Z_{\mathrm{n}}^2$; the greatest-degree term of the final norm is a_2^4. These observations justify equality, rather than just an upper bound, through all potentially cancelling subtractions.

As a consistency check on the retained finalizer algebra, inserting the old source degrees

$$x = 120t + 24, \qquad y = 80t + 16, \qquad c = 4t + 1$$

into $456x + 896y + 24c + 1760$ gives $126496t + 27064$, exactly package 31's degree formula.

C Attribution and integration notes

The new contribution is the direct quadratic-system source, its explicit Sidon instance, the tightened multiple-target support argument, the complete new circuit and degree calculation, and the accompanying checks. The K215 machine, native compiler, parent system and finalizer were obtained from ProveIt at the stated pin. The coefficient builder and DAG utility retain their upstream provenance. Sun's variable-count theorem, Lucas identities, binomial bridge, and residue interpolation are explicitly inherited. Classical Sidon constructions supply the weight design.

This is an AI-assisted research package, not a claim of endorsement by the cited authors or by the repository maintainers. Original licenses and machine notices are included. The proposed directory name `38-quadratic-packing` is an integration suggestion only; no repository commit, publication, or upstream modification was made.

For review, the shortest useful path is: read Lemma 3.1; inspect the arbitrary-g bounds in Section 4; check that the parametric bridge uses precisely Proposition 4.1; and run the supplied verifier. The separate receiver candidate should remain marked incomplete until its stated alignment obligation is proved.

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